Module-Wrestler Online Character Dictionary

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Abstract

We are the most modern parody of a long-established dictionary. We relate the language of characters with the language of finite-dimensional complex group modules.

Contents

1 The Translation Dictionary **2**
- 1.1 Definitions in the Language of Characters 2
- 1.2 Grammatical Rules of the Language of Characters 2
- 1.3 Translating Between Characters and Representations 3

2 Proofs **5**

3 Some Character Tables **6**

1 The Translation Dictionary

There are categories **FinGroup** of finite groups, **FinVect** of finite-dimensional vector spaces over $\mathbb{C}$, and for each $G : \mathbf{FinGroup}$, a category $\mathbf{FinVect}^G$ of finite-dimensional complex G-modules. If $V : \mathbf{FinVect}^G$, the G-action will be denoted $G \xrightarrow{\rho_V} \mathrm{GL}(V)$. The set of conjugacy classes will be denoted $\mathcal{CC}(G)$. For $x \in G$, its conjugacy class is denoted $\mathcal{CL}(x)$.

1.1 Definitions in the Language of Characters

Throughout, $G : \mathbf{FinGroup}$ is a group and $V : \mathbf{FinVect}^G$ is a finite-dimensional complex G-module. (Any other type data will be specified.)

(D1) **Character** of V: The function $\chi_V := \mathrm{tr} \circ \rho_V : G \to \mathbb{C}$. It is a class function (D3), so it is often factored through the quotient $G \twoheadrightarrow \mathcal{CC}(G)$, where it is regarded as a map $\mathcal{CC}(G) \to \mathbb{C}$.

(D2) **Character table** of G: A listing of the characters χ_I corresponding to the finite-dimensional complex irreducible modules I of G. As characters are class functions (D3), the value on each *conjugacy class* is listed. As the irreducible modules biject with the conjugacy classes, this is a square matrix. For example, behold the character table of S_2 (it's not interesting):

Character	$(*)(*)$	$(**)$
χ_{trivial}	1	1
χ_{sign}	1	-1

Beware – although the row for χ_I encodes the character χ_I, it is missing information on the size of the conjugacy class. This information is critical to computing the inner product of two characters (D5); generically, the inner product **does not** agree with the "dot product".

(D3) **Class function**: A function $G \to A$ which is constant on conjugacy classes of G. A is typically $\mathbb{C}$.

(D4) **Kernel** of the character χ_V: The normal subgroup $\{g \in G \mid \chi_V(g) = \chi_V(1)\}$. More details at (T8).

(D5) **Inner product** on $\mathbb{C}^G$: The vector space of functions $G \to \mathbb{C}$ carries Hermitian inner product $\langle -, - \rangle$ given by

$$\langle \varphi, \psi \rangle := \frac{1}{|G|} \sum_{g \in G} \varphi(g) \cdot \overline{\psi(g)}$$

or, for characters $(U, V : \mathbf{FinVect}^G)$:

$$\langle \chi_U, \chi_V \rangle := \frac{1}{|G|} \sum_{g \in G} \chi_U(g) \cdot \chi_V(g^{-1})$$

because of (G3).

1.2 Grammatical Rules of the Language of Characters

I mean in this section to list some key properties that characters always satisfy. If you were to communicate using characters, these are some rules that "always appear" in your sentences! Throughout, $G : \mathbf{FinGroup}$ and $U, V : \mathbf{FinVect}^G$ unless otherwise specified. The set of conjugacy classes is denoted $\mathcal{CC}(G)$, and $\mathcal{I}(G)$ denotes the set of irreducible finite-dimensional complex G-modules, listed once per isomorphism class. Conjugacy classes are denoted $\mathcal{CL}(g)$ and centralisers by $\mathcal{CN}(g)$.

(G1) **Column orthogonality of character tables**: Listing a character table with each character along the rows and each conjugacy class along the columns,

$$X := \begin{array}{c} \\ \chi_{\text{trivial}} \\ \chi_{I_1} \\ \vdots \\ \chi_{I_{k-1}} \end{array} \begin{array}{c} \begin{array}{cccc} \mathcal{CL}(1_G) & \mathcal{CL}(g_1) & \cdots & \mathcal{CL}(g_{k-1}) \end{array} \\ \begin{pmatrix} * & * & \cdots & * \\ * & * & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \cdots & * \end{pmatrix} \end{array}$$

we have that each column is orthogonal under the usual dot product[1]. That is, for $g, h \in G$ with distinct conjugacy classes $\mathcal{CL}(g) \neq \mathcal{CL}(h)$,

$$\sum_{i=0}^{k-1} \chi_{I_i}(g)\overline{\chi_{I_i}(h)} = 0$$

whilst to give the square-norm of a column, we have

$$\sum_{i=0}^{k-1} |\chi_{I_i}(g)|^2 = \frac{|G|}{|\mathcal{CL}(g)|} = |\mathcal{CN}(G)|.$$

(G2) **Dimensions/degrees of irreducibles divide centre indices**:

$$\dim(V) = \chi_V(1) \mid [G : Z(G)].$$

(G3) **Inverses**: For $g \in G$, $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

(G4) **Kernels are normal**: $\operatorname{Ker} \chi_V$ is a normal subgroup of G.

(G5) **Frobenius reciprocity**: If $H \leq G$, $U : \mathbf{FinVect}^H$ and $V : \mathbf{FinVect}^G$, then

$$\langle \operatorname{Ind}_H^G U, V \rangle = \langle U, \operatorname{Res}_H^G V \rangle.$$

See (T2).

(G6) **Orthonormality of irreducibles**:

$$U, V \text{ both irreducible} \implies \langle \chi_U, \chi_V \rangle = \mathbf{1}_{U \simeq V} \quad \text{(indicator function)}.$$

(G7) **Orthonormal basis formed by irreducibles**: The vector space $\mathbb{C}^{\mathcal{CC}(G)}$ carries orthonormal basis $\{\chi_I \mid I \in \mathcal{I}(G)\}$. The inner product is **not** the "dot product", but rather as defined at (D5).

1.3 Translating Between Characters and Representations

Throughout, $G : \mathbf{FinGroup}$ and $U, V : \mathbf{FinVect}^G$.

(T1) **Abelian**: The group G is Abelian just when its character table is $|G| \times |G|$, itself just when each irreducible character is one-dimensional.

(T2) **Adjunction of induction to restriction**: For a subgroup $H \leq G$, the adjunction of functors $\operatorname{Ind}_H^G \dashv \operatorname{Res}_H^G$ appears as **Frobenius reciprocity**:

$$\mathbf{FinVect}^G(\operatorname{Ind}_H^G U, V) \simeq_{\text{nat.}U,V} \mathbf{FinVect}^H(U, \operatorname{Res}_H^G V) \quad \rightsquigarrow \quad \langle \operatorname{Ind}_H^G U, V \rangle = \langle U, \operatorname{Res}_H^G V \rangle.$$

The inner product relation is a decategorification of the (categorical) adjunction.

[1] This inner product is *not* weighted by the sizes of the conjugacy classes.

(T3) **Commutator**: The commutator subgroup $[G, G]$ is the intersection of all one-dimensional characters.

$$[G, G] = \bigcap_{I \in \mathcal{I}(G).\, \dim I = 1} \operatorname{Ker} \chi_I.$$

(T4) **Dimension/Degree** of V:

$$\dim(V) = \chi_V(1).$$

(T5) **Dimension of hom-space** $\mathbf{FinVect}^G(U, V)$ (this space often goes by $\operatorname{Hom}_G(U, V)$):

$$\dim \mathbf{FinVect}^G(U, V) = \langle \chi_U, \chi_V \rangle.$$

(T6) **Equivalence of irreducible modules**: In the context where U, V are irreducible,

$$U \simeq V \iff \langle \chi_U, \chi_V \rangle = 1$$

and furthermore, in the same context,

$$U \not\simeq V \iff \langle \chi_U, \chi_V \rangle = 0.$$

See (G6) for a more concise statement.

(T7) **Equivalence of modules**:

$$U \simeq V \iff \chi_U = \chi_V.$$

(T8) **Normal subgroups**: For $N \subseteq G$,

$$N \trianglelefteq G \iff \exists \mathcal{J} \subseteq \mathcal{I}(G).\ N = \bigcap_{I \in \mathcal{J}} \operatorname{Ker} \chi_I.$$

(T9) **Frobenius reciprocity**: See (T2).

(T10) **Irreducible factors**: For U irreducible,

$$\text{a factor of } U \text{ appears in } V \iff \langle \chi_U, \chi_V \rangle \neq 0.$$

(A factor "appearing" means that $V \simeq U \oplus (\text{stuff})$.)

(T11) **Irreducibility**: See also (G6).

$$V \text{ is irreducible} \iff \langle \chi_V, \chi_V \rangle = 1.$$

(T12) **Order of centralisers**: For $g \in G$,

$$|\mathcal{CN}(g)| = \sum_{I \in \mathcal{I}(G)} |\chi_I(g)|^2.$$

(T13) **Order of group**:

$$|G| = \sum_{I \in \mathcal{I}(G)} |\chi_I(1_G)|^2 = \sum_{I \in \mathcal{I}(G)} (\dim I)^2.$$

2 Proofs

All proofs are collected here. They are all reproduced by myself for my own practice. FIXME: Actually do it

3 Some Character Tables

Here are some character tables. FIXME: Actually do it